Abstract

This study evaluates a simple retrieval-augmented generation system for answering questions from documents. The system uses parent and child chunks to improve retrieval accuracy and preserve context.

Introduction

Retrieval-augmented generation retrieves relevant information from external documents before generating an answer. Effective chunking is important because chunks that are too large may contain unnecessary information, while chunks that are too small may lose context.

Methodology

Documents are divided into semantic parent sections and smaller child chunks. Each child contains approximately 150 to 200 tokens with a small overlap between neighboring chunks. Child embeddings are stored in a vector database for retrieval.

Results and Discussion

The smaller child chunks provide more precise retrieval results. Neighboring children can be added after retrieval to provide additional context without sending the entire parent to the language model.

Conclusion

Parent-child chunking can improve retrieval precision while reducing unnecessary context. A combination of semantic parents, small children, and neighbor expansion provides a practical RAG strategy.

References

Lewis et al. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.